



Elier Garcia

ELECTRICAL ENGINEER · INDUSTRIAL
INSTALLATIONS

Contact

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Skills

Electrical Engineering

AutoCAD · OPUS · LV/MV Installations ·
NEC/IEC/NOM Standards · Site
supervision · Subcontractor management
· Office 365

Backend & Cloud

Supabase · PostgreSQL · Docker (Self-
hosted)

AI / ML & Data

Gemini AI · RAG Architectures · pgvector
· Python · SQL

DevOps & Tooling

n8n · Power BI

Languages

Spanish (native)
English B2
German (basic)

Awards & certifications

Electric Power Systems, University at
Buffalo (SUNY) · February 2023

PVC-Coated Conduit Installer, Plasti-
Bond · April 2024

**Artificial Intelligence: Principles &
Applications**, Universidad del Valle de
México · August 2026

Digital Leadership Programme, INESDI
Business TechSchool · April 2025

Profile

Electrical engineer with two years at DEMEK on large-scale industrial projects: one year estimating electrical installations and one on site, first at Microsoft's hyperscale data center in Querétaro, where I went from newest resident to sole lead of the outdoor front, and then at BMW Group's plant in San Luis Potosí. M.S. in Business Analytics & AI with the top record of my cohort: I apply data and automation to the problems I lived on site, such as materials tracking and technical documentation.

Work experience

Founder · AI & Automation Engineer

2026 — present

EG Solutions · Madrid (remote)

- Design and development of web platforms, n8n automations and AI agents for businesses.

Electrical Construction Resident

Jan 2024 — Feb 2025

DEMEK S.A. de C.V. · Querétaro / San Luis Potosí, Mexico

- **Microsoft Azure hyperscale Data Center** (Querétaro, 9 months): from newest resident to **sole lead of the outdoor electrical front** in 6 months, directly supervising **60+ people**, with daily progress meetings, weekly planning and coordination with mechanical, telecom and civil works.
- Outdoor scope: medium voltage (conduit and cabling, coordinated with civil works), grounding systems, pump room (MCC, lighting and power panels), outdoor pole lighting and temporary site power.
- Installation, interconnection and quality verification of critical power equipment: diesel generators, modular UPS, dry-type and oil-filled outdoor transformers, MCCs, switchboards and PDUs, with crane coordination and redundant A/B power cabling.
- **BMW Group plant** (San Luis Potosí, 5 months, 19,847 m² body-shop expansion in 8 sections): supervised LV and grounding in 2 sections (~40 people) and handled material requisitions for nearly all 8.
- Built my own materials tracking system (Excel + SAP): every off-budget request technically justified (take-off vs. drawings or documented design change), fewer supply stoppages, and irregularities detected that led to safety improvements.
- Compliance with NEC/IEC/NOM specifications and safety standards on both sites.

Electrical Budget Analyst

Jan 2023 — Jan 2024

DEMEK S.A. de C.V. · Chihuahua, Mexico

- **30+ bids delivered, up to USD 500K**, several in parallel (1-2 week deadlines): drawings and specs in AutoCAD, Excel quantity take-offs by room, system and area, and OPUS pricing by line item.
- Supplier quotes under client-defined brand and function specs, full indirect costs (security, site offices, cranes, labor, safety and documentation) and change orders during execution.
- Won the **Terex project in Nuevo León**: the only bid that reached the final presentation, delivered in English directly to Terex through Copachisa.

Related project

PDF Technical Assistant

pdf-tech-assistant2.vercel.app

Built from my own site experience to query electrical specs and technical manuals: each company uploads its PDFs and gets answers only from the documents, focused on its specialty (electrical, civil, mechanical...). Multi-company with roles, delivered as SaaS or on-premise with Docker.

React 18 · TypeScript · Supabase Edge Functions · pgvector · Gemini

Education

B.S. in Electrical Engineering

Instituto Tecnológico de Chihuahua · Mexico · 2023

Major in electrical installations. GPA 9.27/10. President of the Electrical Engineering student committee.

M.S. in Business Analytics & Artificial Intelligence

INESDI Business TechSchool · Madrid, Spain · 2026

Top academic record of the cohort (8.94/10). ETL/ELT, RAG architectures, applied machine learning and Power BI.